

The DNA GOLDRUSH

India's hospital chains, diagnostics networks, and global investors have poured millions of dollars into gene sequencing, bioinformatics and population-scale genomic capacity to bring DNA testing to India. There is one glitch: the cost remains prohibitive

BY **NEETU CHANDRA SHARMA**



THE GENOMIC REPORT arrives on Dr. Mandeep Singh Malhotra's desk as early morning sun fills the oncology wing at New Delhi's CK Birla Hospital.

It is 7:40 am, minutes before the outpatient department opens, and he knows the contents of the envelope could alter the patient's treatment pathway.

Inside the envelope is a multi-gene tumour sequencing report, one of the thousands being processed each week in India. The report will determine whether targeted therapy is an option or whether chemotherapy should continue.

"There is no room for guesswork anymore," says Dr. Malhotra, his eyes darting through the variants, amplifications and deletions suggested in the report.

A few years ago, this test would have taken weeks, cost many times what it does now, and required samples to be shipped to labs abroad. That has changed. India's hospital chains, diagnostics networks and global investors have poured hundreds of millions of dollars into gene sequencing, bioinformatics and population-scale genomic capacity to bring DNA testing to the doorstep of Indians.

Clinical decisions, insurance approvals and oncology pathways are now moving in step with a new architecture. Specialists like Dr Malhotra, who is Director of Surgical Oncology at CK Birla Hospital, now treat genetic information as essential.

A genomic report details findings from genetic analysis, identifying DNA variations, crucial for diagnosing diseases, predicting prognosis, and guiding therapies, especially in cancer.

The testing process begins with a sample usually extracted from a blood or tissue in a tumour. Genomic testing works by reading a patient's DNA. A lab machine, called a sequencer, "reads" the vast chemical library, creating billions of data points.

Experts then use advanced software and artificial intelligence